BA820 – Project M2

Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Bhavya Bavishi

1. Refined Problem Statement & Focus

Question: How do age-gap distributions differ across couple types in Hollywood films, and are same-gender pairings closer to age parity?

Key shift from M1 to M2: The initial milestone made one thing immediately clear: Hollywood overwhelmingly relies on traditional man–woman couples. Over 80% of the dataset falls into this category, while same-gender couples appear fewer than 25 times combined. From both a statistical and a business perspective, treating these groups as equal would lead to weak and potentially misleading conclusions.

Rather than forcing comparisons the data cannot support, this project refocused on a more meaningful and actionable distinction: within mixed-gender couples, does it matter which partner is older?

The answer is yes—and the difference is not subtle.

Films with older women and younger men show a median age gap of about 3 years, while films with older men and younger women show a median age gap of about 10 years. That seven-year difference is larger than almost any other contrast in the dataset and represents a fundamentally different casting norm.

Conceptual shift: Instead of asking, "Are average age gaps different?", this project reframes the question in business terms: *Do different couple types follow different casting logics that shape on-screen age inequality?*

This reframing allows the analysis to focus on patterns, norms, and regimes of behavior, rather than isolated statistics.

2. EDA & Preprocessing: Updates

Where M1 left off: Age gaps in Hollywood are highly uneven. While many couples are relatively close in age, a meaningful minority feature very large gaps—sometimes exceeding 50 years. Early visual analysis suggested gender dynamics might play a role, but M1 did not formally establish whether those patterns were stable or meaningful.

What I added in M2:

- **Testing what "age parity" really means:**
  Instead of relying on a single cutoff, I tested multiple reasonable definitions of age parity (e.g., within 3 years and within 5 years). Importantly, the same conclusions held across both definitions. This suggests that the findings are not driven by arbitrary choices, which strengthens their credibility.
- **Introducing a simple business proxy for star power:**
  To explore whether age gaps might be influenced by market dynamics rather than norms alone, I used actor recurrence as a rough indicator of prominence. While imperfect, this proxy captures a real industry intuition: actors who appear repeatedly tend to have more leverage in casting decisions.
- The relationship is modest (Spearman correlation ≈ 0.14), but statistically meaningful. In practical terms, this suggests that star power may enable larger age gaps, even if it does not fully explain them.
- **Moving from single metrics to behavioral profiles:**
  Rather than comparing couple types using only averages, I constructed richer profiles that capture typical gaps, variability, frequency of extreme cases, and how patterns shift over time. This allows the analysis to treat casting behavior as a system, not just a number.
- **Clarifying couple-type definitions:**
  The dataset lists the older partner first. This means "man–woman" explicitly refers to an older man paired with a younger woman, while "woman–man" indicates the reverse. This distinction is central to interpretation and business relevance.

No additional data cleaning was required. The dataset is well-maintained; the analytical effort focused on making the insights clearer, more robust, and more interpretable.

3. Analysis & Experiments

<u>Method 1: Identifying Casting Regimes</u>

Business Motivation:

Executives rarely think in terms of averages. They think in categories—what's typical, what's risky, and what's extreme. To align with that mindset, I examined whether age gaps naturally fall into a small number of recognizable casting "regimes."

What Emerged:

Four distinct regimes appeared, ranging from near-age parity to extreme age gaps. Most films cluster in the middle, but a noticeable share consistently pushes toward much larger gaps.

Key Insight:

- Films where the woman is older overwhelmingly fall into the near-parity regime. About 72% of these couples land there.
- By contrast, films where the man is older are spread across all regimes—including the most extreme—indicating far greater tolerance for age inequality.
- This is not just a difference in degree; it reflects two different casting logics operating within the same industry.

Unexpected Observation:
Same-gender male couples show a relatively high share of extreme age gaps. However, because this group contains only 12 observations, this finding should be viewed as directional rather than definitive.

<u>Method 2: Comparing Overall Casting Profiles</u>
Business Motivation:
If a couple of types truly behave differently, their overall patterns should separate cleanly when viewed holistically.

What the Results Show:
When all behavioral features are viewed together, woman–man couples clearly separate from all others. Interestingly, man–woman, man–man, and woman–woman couples cluster together—even though their sample sizes differ dramatically.

Interpretation:
When a man is older, Hollywood casting looks broadly similar regardless of partner gender. When a woman is older, the rules change.

Independent clustering methods confirm this split, reinforcing that the pattern reflects structural behavior, not noise.

<u>Being Explicit About Uncertainty</u>
From a business standpoint, understanding where conclusions are strong—and where they are not—is critical.

- **Same-gender couples:** With sample sizes under 15, uncertainty is unavoidable. Confidence intervals for parity rates span nearly 40 percentage points, making firm conclusions impossible.
- **Mixed-gender directionality:** The difference between older-man and older-woman pairings is both large and statistically robust (permutation $p \approx 0.0001$). This is the most reliable result in the study.

Bottom Line: The data strongly support a gender-directional effect in mixed-gender couples. Other comparisons should be treated as exploratory.

4. Findings & Interpretations

Finding 1: Age inequality is directional, not compositional

When women are older than their male partners, age-balanced casting is common (over 50% near parity). When men are older—regardless of partner gender—large age gaps are much more frequent.

Business Meaning: Hollywood's age-gap issue is less about the type of couple and more about which gender is allowed to age on screen.

Finding 2: Same-gender parity remains unresolved.

Same-gender couples do not clearly eliminate age gaps, but the evidence is too limited to draw strong conclusions.

Business Meaning: Representation alone does not automatically resolve structural inequalities.

Finding 3: Woman–man couples follow a distinct casting logic

These films overwhelmingly occupy age-balanced regimes, while man–woman films span the full spectrum—including extremes.

Actionable Insight: Studios seeking to reduce age inequality can track not just average gaps, but the share of projects that fall into high-inequality regimes.

Finding 4: Star power plays a supporting role

Repeatedly cast actors are modestly associated with larger age gaps. While this does not prove causation, it aligns with known industry dynamics around leverage and bankability.

5. Next Steps

Planned extensions include:

1. Examining how casting regimes change over time
2. Incorporating genre and narrative context
3. Using external measures of star power to better test economic explanations

M3 Direction:

Apply time-based analysis, integrate richer external data, and use hierarchical models that allow sparse groups to borrow information from the broader system.

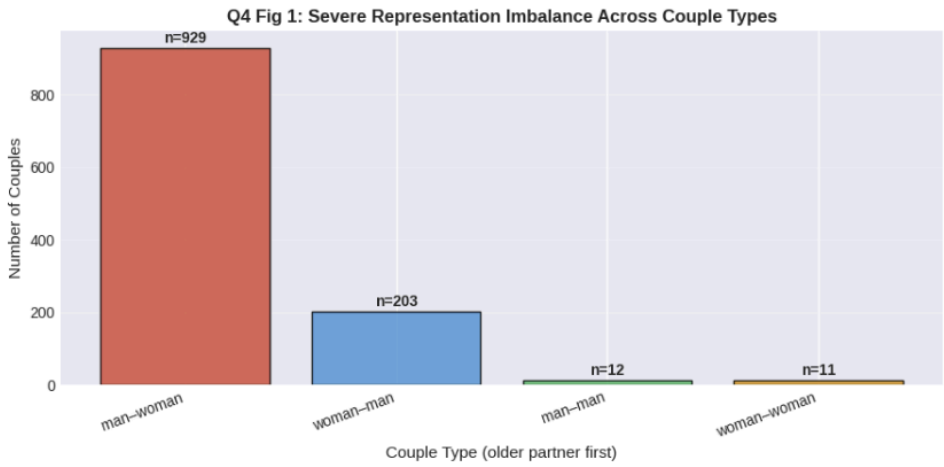
Appendix

Shared GitHub Repository (Required)

- https://github.com/patelparisha/ba820-b1-team10
- My files exist in the Bhavya Bhavishi named folder in the Milestones branch of the git repository.

Supplemental Material (Highly Recommended)

Figure 1:



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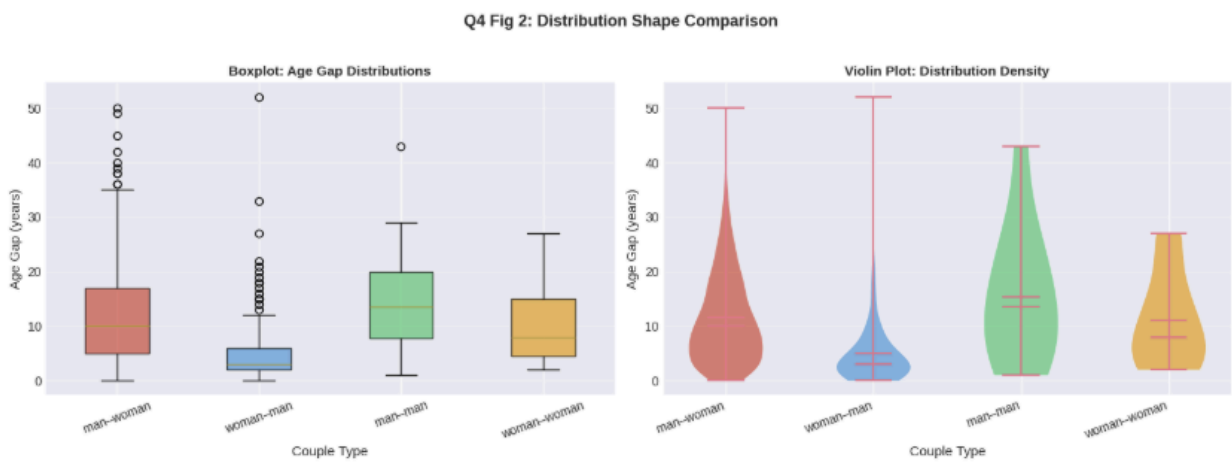

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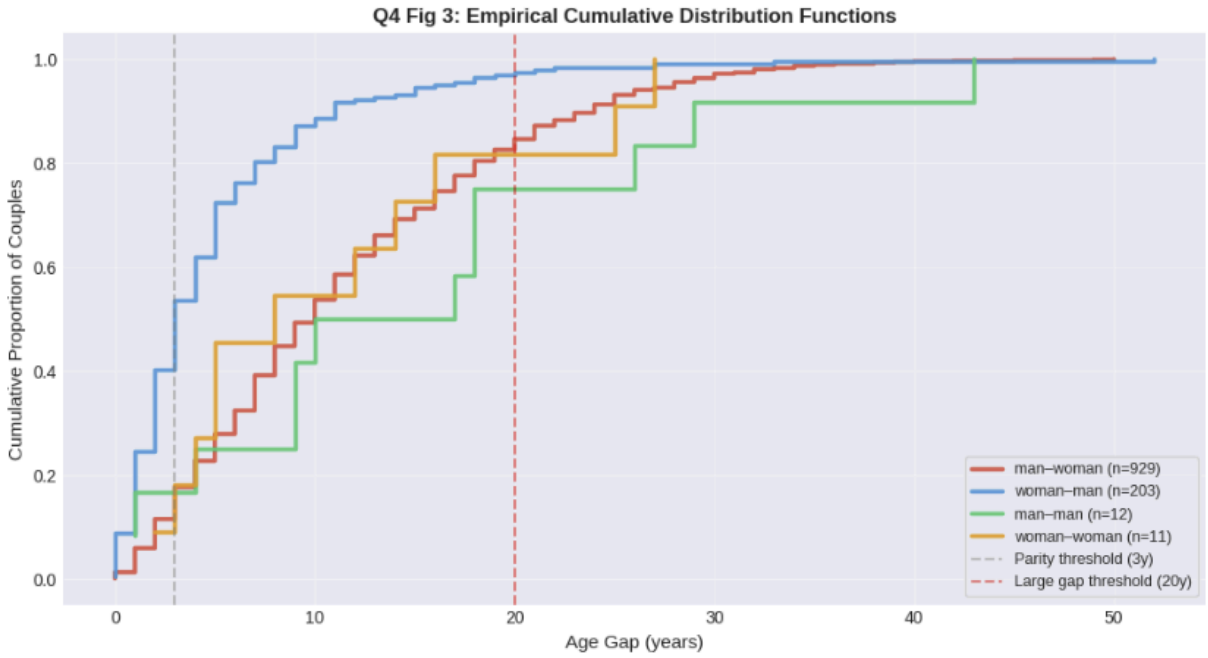

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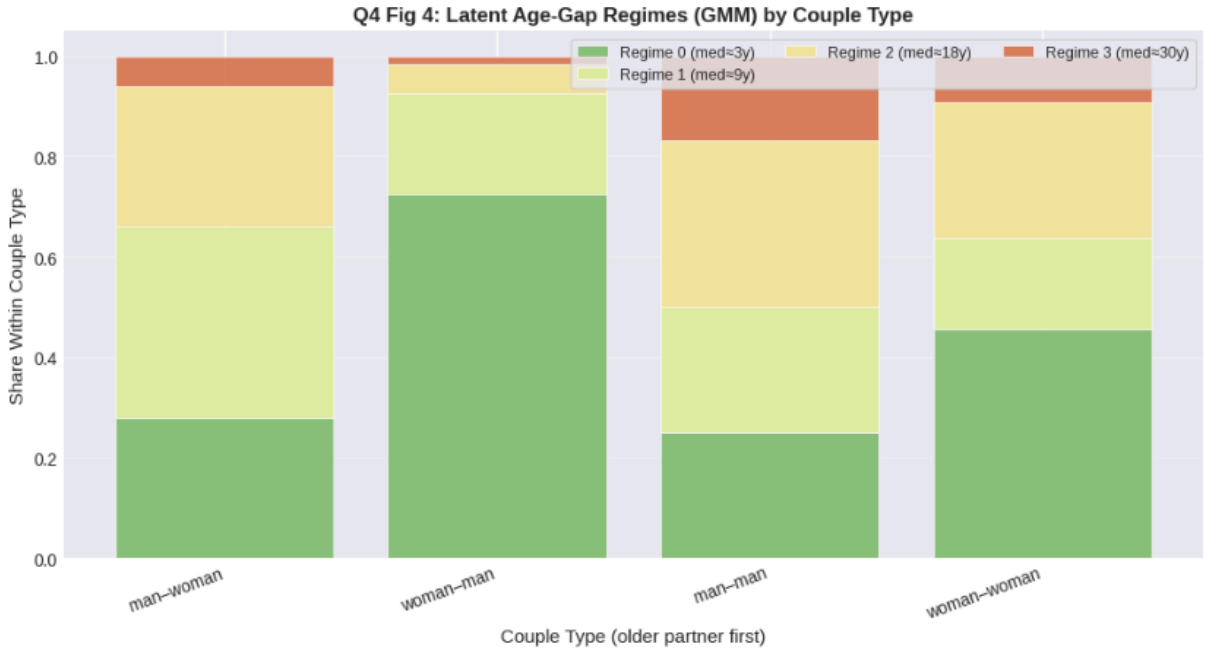

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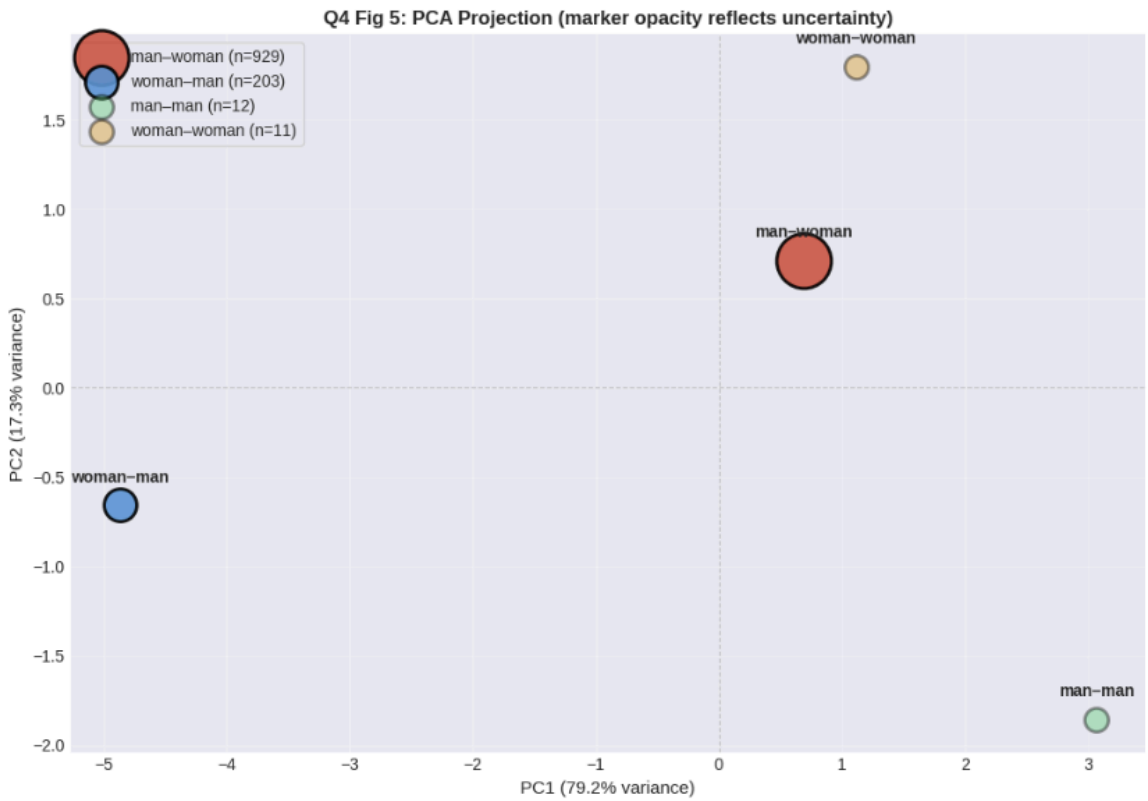

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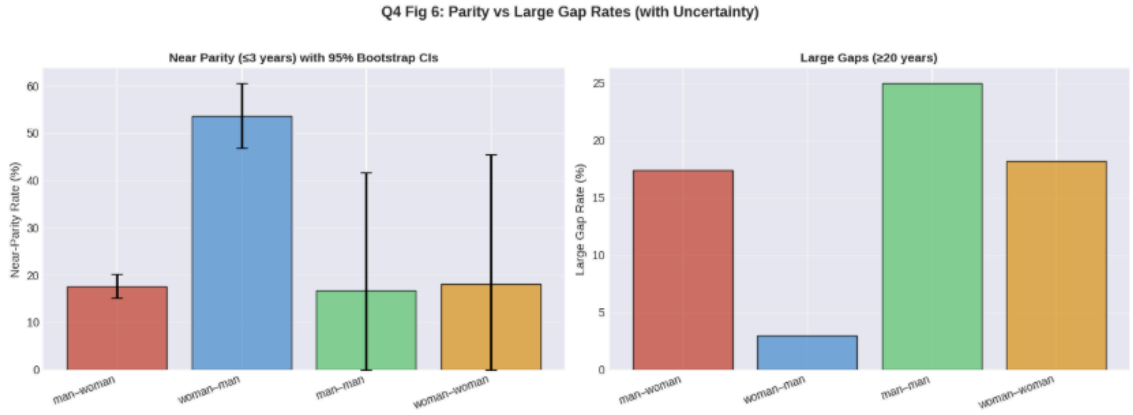

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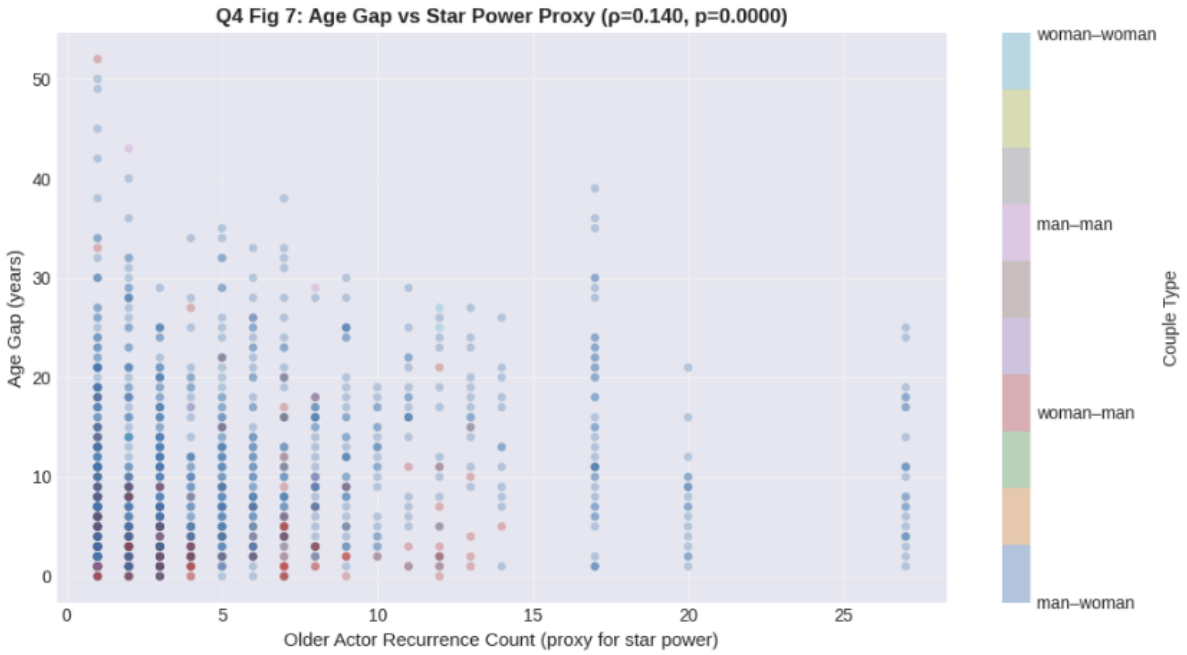

Process Overview

The analytical process began by working with the Hollywood Age Gap dataset and preparing it for structured analysis of director behavior. Initial steps focused on organizing the available information into meaningful indicators of casting patterns and collaboration tendencies, ensuring that the dataset could support stable comparisons across directors. This included summarizing relationship patterns between actors and directors and constructing consistent profiles that represent how directors typically cast on-screen couples.

Once these profiles were established, the analysis shifted toward identifying broader patterns across directors. Grouping techniques were used to uncover distinct casting "regimes," while visualization methods helped interpret how directors differed from one another along key dimensions. To ensure the findings were reliable, multiple validation checks were conducted to confirm that the observed patterns were not driven by chance or small-sample effects.

Finally, the results were consolidated into a set of visual summaries and interpretable insights that connect casting behavior to industry dynamics. This process created a foundation for understanding recurring casting structures in Hollywood and set up future work exploring how these patterns evolve over time and across collaboration networks.

Use of Generative AI Tools

1. https://claude.ai/share/06fe3e74-3873-4d75-9070-4b58e52a678d